PGCP-AC AUGUST 26
Assignment No- 4

Write Create a Employee ,Department & Project Tables .Add atleast 5 records in each table & Write a following Queries.

1. Write a Relational Algebra expression to perform **Cartesian Product (Cross Join)** between Emp and Dept tables.

ANS -> SELECT * FROM EMP CROSS JOIN DEPT;

2. Write a SQL query to perform an **INNER JOIN** between Emp and Dept tables displaying EmpName and DeptName.

ANS -> SELECT EMP.EmpName, DEPT.DeptName FROM EMP INNER JOIN DEPT ON EMP.DeptID = DEPT.DeptID;

3. Write a SQL query to perform a **LEFT OUTER JOIN** between Emp and Dept tables to display all employees along with their department information, including employees without a department.

ANS -> SELECT * FROM EMP LEFT OUTER JOIN DEPT ON EMP.DeptID = DEPT.DeptID;

4. Write a SQL query to perform a **RIGHT OUTER JOIN** between Emp and Dept tables.

ANS-> SELECT * FROM EMP RIGHT JOIN DEPT ON EMP.DeptID = DEPT.DeptID;

5. Write a SQL query to perform a **NATURAL JOIN** between Emp and Dept.

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ANS -> SELECT * FROM EMP NATURAL JOIN DEPT;

6. Write a SQL query to perform a **CROSS JOIN** between Emp and Project table.

ANS -> SELECT * FROM EMP CROSS JOIN PROJECT;

7. Write a SQL query to **create a new table EmpBackup with the same structure as Emp but no data**.

ANS -> CREATE TABLE EmpBackup LIKE Emp;

8. Write a SQL query to **copy all data from Emp into EmpBackup table**.

ANS -> INSERT INTO EmpBackup SELECT * FROM Emp;

9. Write a SQL query to **create a new table ProjectArchive with the same structure and data as Project**.

ANS -> CREATE TABLE ProjectArchive AS SELECT * FROM Project;

10. Write a SQL query to create an **AUTO_INCREMENT sequence for EmpID in the Emp table** during table creation.

ANS -> CREATE TABLE EMP (EmpID INT PRIMARY KEY AUTO_INCREMENT);

11. Write a SQL query using a **subquery** to find all employees whose salary is greater than the **average salary of all employees**.

ANS -> SELECT * FROM EMP WHERE Salary > (SELECT AVG(Salary) FROM EMP);

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12. Write a SQL query using a **subquery in WHERE clause** to find employees working in departments located in 'Mumbai'.

ANS -> SELECT * FROM EMP WHERE DeptID = ANY (SELECT DeptID FROM DEPT WHERE Location = 'Mumbai');

13. Write a SQL query to display all departments where the number of employees is greater than **5**, using a subquery.

ANS -> SELECT * FROM DEPT WHERE DeptID = ANY (SELECT DeptID FROM EMP GROUP BY DeptID HAVING COUNT(*) > 5);

14. Write a subqueries for find out Max salary of employee.

ANS -> SELECT * FROM EMP WHERE Salary = ANY (SELECT MAX(Salary) FROM EMP);

15. Write a self join query for each table.

ANS ->

SELECT E1.EmpName AS 'Employee Name', E2.EmpName AS 'Manager Name' FROM EMP AS E1 JOIN EMP AS E2 ON E1.ManagerID = E2.EmpID;

SELECT D1.DeptName AS 'Parent Department Name', D2.DeptName AS Department Name' FROM DEPT AS D1 JOIN DEPT AS D2 ON D2.ParentDeptID = D1.DeptID;

SELECT P1.ProjectName AS 'Related Project Name', P2.ProjectName AS 'Project Name' FROM PROJECT AS P1 JOIN PROJECT AS P2 ON P2.RelatedProjectID = P1.ProjectID;

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